Test Paper-14

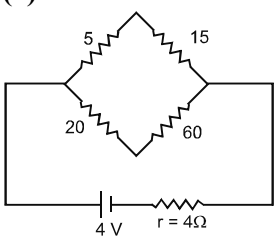
ANSWER KEY

1.	**(2)**	**16.**	**(2)**	**31.**	**(2)**	**46.**	**(89)**	**61.**	**(1)**
2.	**(1)**	**17.**	**(3)**	**32.**	**(2)**	**47.**	**(15)**	**62.**	**(2)**
3.	**(4)**	**18.**	**(2)**	**33.**	**(4)**	**48.**	**(3)**	**63.**	**(4)**
4.	**(3)**	**19.**	**(1)**	**34.**	**(4)**	**49.**	**(4)**	**64.**	**(2)**
5.	**(4)**	**20.**	**(3)**	**35.**	**(4)**	**50.**	**(1)**	**65.**	**(1)**
6.	**(3)**	**21.**	**(50)**	**36.**	**(4)**	**51.**	**(1)**	**66.**	**(3)**
7.	**(1)**	**22.**	**(3)**	**37.**	**(4)**	**52.**	**(1)**	**67.**	**(4)**
8.	**(4)**	**23.**	**(5)**	**38.**	**(1)**	**53.**	**(4)**	**68.**	**(4)**
9.	**(3)**	**24.**	**(3)**	**39.**	**(1)**	**54.**	**(4)**	**69.**	**(4)**
10.	**(2)**	**25.**	**(18)**	**40.**	**(2)**	**55.**	**(2)**	**70.**	**(3)**
11.	**(3)**	**26.**	**(1)**	**41.**	**(1)**	**56.**	**(1)**	**71.**	**(3)**
12.	**(4)**	**27.**	**(3)**	**42.**	**(1)**	**57.**	**(2)**	**72.**	**(260)**
13.	**(2)**	**28.**	**(2)**	**43.**	**(2)**	**58.**	**(2)**	**73.**	**(5)**
14.	**(2)**	**29.**	**(2)**	**44.**	**(1)**	**59.**	**(1)**	**74.**	**(3)**
15.	**(1)**	**30.**	**(1)**	**45.**	**(3)**	**60.**	**(1)**	**75.**	**(6)**

SOLUTIONS

PHYSICS

1. (2)



$$\Rightarrow R_{eq} = \frac{80 \times 20}{80 + 20} + 4 = 20\Omega$$

$$\Rightarrow I = \frac{V}{R} = \frac{4}{20} = \frac{1}{5}\text{ A}$$

2. (1)

Let ρ be the density of the material. ρ_0 be the density of water. When the sphere has just sunk in water, then

Weight of the sphere = Weight of the water displaced

$$\Rightarrow \frac{4}{3}\pi\left(R^3 - r^3\right)\rho_s g = \frac{4}{3}\pi R^3 \rho_w g$$

$$\Rightarrow (R^3 - r^3)\rho_s = R^3\rho_w$$

$$\Rightarrow \frac{\left(R^3 - r^3\right)}{R^3} = \frac{\rho_w}{\rho_s}$$

$$\Rightarrow 1 - \left(\frac{r}{R}\right)^3 = \frac{1}{8}$$

$$\Rightarrow \frac{r}{R} = \frac{7^{1/3}}{2}$$

3. (4)

Process$_{1\to2}$ and Process$_{3\to4}$ are isochoric process.

$W_{12} = 0$

$W_{34} = 0$

$W_{23} = n\,R\,(T_3 - T_2) = 3\,R\,(2400–800) = 4800\,R$

$W_{41} = nR\,(T_1 - T_4) = 3\,R\,(400–1200) = –2400\,R$

$W = (4800 – 2400)\,R = 2400\,R = 20$ kJ

4. (3)

Electric charge in uniform motion produces both electric magnetic fields

5. (4)

When a P-N junction diode is formed an electric field is developed from N-side to P-side due to which barrier potential is created & majority charge carrier can not flow through the junction due to barrier potential so current is zero unless we apply forward bias voltage.

6. **(3)**

Spontaneous fission occurs to lower the binding energy of product nuclei. So statement 1 is true and statement 2 is false.

7. **(1)**

As magnetic moment is a vector, so when magnets are joined together pole for pole, the resultant magnetic moment is $M + 2M = 3M$; so

$$T = 2\pi\sqrt{\frac{(I_1 + I_2)}{3MB}} \qquad \ldots\ldots(1)$$

When the polarity of one of the magnets is reversed, $M' = M \sim 2M = M$;so

$$T' = 2\pi\sqrt{\frac{(I_1 + I_2)}{MB}} \qquad \ldots..(2)$$

Dividing eqn. (2) by (1)

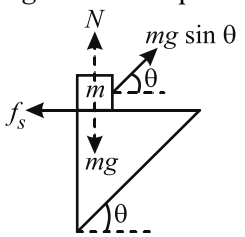
$$\frac{T'}{T} = \sqrt{3},$$

i.e. $T' = (\sqrt{3})T = 3\sqrt{3}$ sec

8. **(4)**

As the block does not slip on prism, the combined acceleration of the prism is $a = g \sin\theta$.

$mg \sin\theta$ is the pseudo force on m.



or $N = mg\cos^2\theta$

And for no slipping, $mg \sin\theta \cos\theta \le \mu N$

$mg\sin\theta\cos\theta \le \mu mg\cos^2\theta$ or $\mu \ge \tan\theta$

9. **(3)**

$C_{eq} = 3C$

$Q_{eq} = 3Q$

$$E = \frac{1}{2}\frac{Q_{eq}}{C_{eq}} = \frac{1}{2}\frac{(3Q)^2}{3C} = \frac{3Q^2}{2C}$$

10. **(2)**

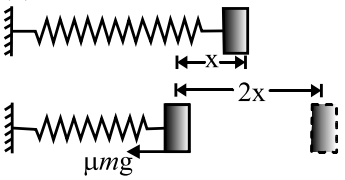
Assertion A is correct: The photoelectric effect does not occur if the energy of the incident radiation is less than the work function of a metal. The work function is the minimum amount of energy required to remove an electron from the metal surface. If the energy of the incident radiation is lower than the work function, the photons do not possess enough energy to overcome the binding energy of the electrons, and no photoelectrons are emitted

$h\nu = w_0 + K.E_{max}$

If $h\nu = w_0$

$\Rightarrow K.E_{max} = 0$

11. **(3)**



PE of spring due to its compression by $x = I$ Work done by frictional force when displaced by $2x$|

i.e., $\frac{1}{2}kx^2 = \mu mg\,2x$

$\Rightarrow \quad x = \frac{4\mu mg}{x}$

$\therefore$ Correct option is (3)

12. **(4)**

Let v and ω be the linear and angular speeds of the rod after applying an impulse J and B. Then from Impulse = change in momentum

We have $mv = J$ or $v = \frac{J}{m}$ …(i)

$$I\omega = J \cdot \frac{l}{2}$$

Or $\frac{ml^2}{12} \cdot \omega = J\frac{l}{2}$ or $\omega = \frac{6J}{ml}$ …(ii)

After the given time $t = \frac{\pi ml}{12J}$, the rod will rotate an angle

$$\theta = \omega t = \left(\frac{6J}{ml}\right)\left(\frac{\pi ml}{12J}\right) = \frac{\pi}{2}$$

$$\omega \times \frac{l}{6} = \left(\frac{l}{6}\right)\left(\frac{6J}{ml}\right) = \frac{J}{m} = v$$

$$\therefore \quad \left|\vec{v}_P\right| = \sqrt{2}v = \sqrt{2}\frac{J}{m}$$

13. **(2)**

$v = \alpha t + \beta t^2$

$\frac{ds}{dt} = \alpha t + \beta t \Rightarrow ds = \left(\alpha t + \beta t^2\right)dt$

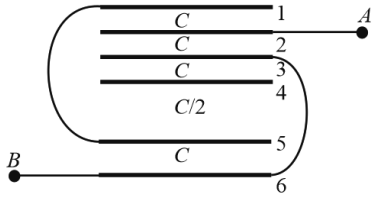
Applying integration both side.

$s = \int_1^2 (\alpha t + bt^2)\,dt$

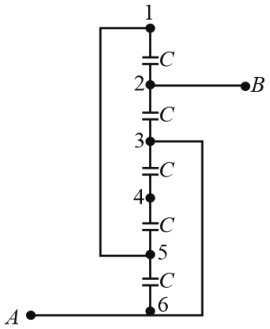
$s = \frac{3}{2}\alpha + \frac{7}{3}\beta$

14. (2)

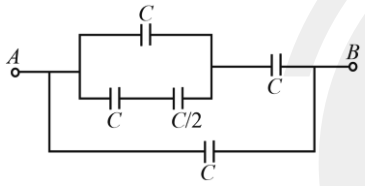
Figure below shows the respective capacitances between adjoining plates numbered 1 to 6



The equivalent circuit of the above system is as shown in figure below



Above circuit can be redrawn as shown below



Thus across terminals AB equivalent capacitance is given as

$$C_{AB} = \frac{11}{7}C = \frac{11}{7}(7\mu F)$$

$$C_{AB} = 11\mu F$$

15. (1)

ρT^2 is a universal constant whose value is $\frac{3\pi}{G}$

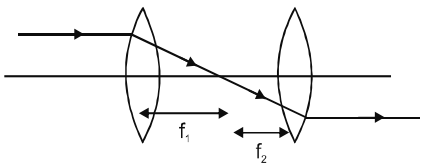
16. (2)

$$\because\ \theta = \frac{\lambda}{a} = \frac{R}{D}$$

and $\lambda_{x\text{-Ray}} < \lambda_{visible}$

$\therefore\ R_{x\text{-Ray}} < 0.1$

17. (3)



Distance between lens is $= f_1 + f_2$,

18. (2)

(I) $P_i = P_f \Rightarrow 0 = 60(1+v) + (80+20)v \Rightarrow v = -\frac{3}{8}$

is opposite to $2\,m/s$ velocity of Ram i.e. $3/8\,m/s$ towards rigid

(II) $0 = 80(1+v) + (60+20)v \Rightarrow v = -\frac{1}{2}$ m/s

(III) $80(1+v) + 60(-1+v) + 20v = 0$

$$80 - 60 + 800 + 60v + 20v = 0$$

$$-20 = 160v \quad v = -\frac{1}{8}\text{ m/s}$$

19. (1)

The wave equation for the wave travelling in negative x direction is given as $y = f(kx + \omega t)$ on the contrary wave travelling in positive x direction is given as $y = f(kx - \omega t)$.

The given equation, $y = \frac{3}{2x + 3t + 4}$ is the from of $y = f(kx + \omega t)$. Hence, the given wave will move in the negative direction of x axis hence, both assertion and reason are true and reason is the correct explanation of assertion.

20. (3)

$$mg\cos\theta = \frac{mv^2}{a}$$

or $$g\cos\theta = \frac{4^2 + 2.g.\frac{a}{4}}{a}$$

or $$g.\frac{3}{4}a = u^2 + 2g\frac{a}{4}$$

$$\Rightarrow \quad u = \frac{\sqrt{ag}}{2}$$

21. (50)

$r = 0.5$ mm $= 0.5 \times 10^{-3}$ m and, $f = 100$ Hz

$T = 100$ N, $v = 100$ m/s

$$v = \sqrt{\left(\frac{T}{\mu}\right)} \Rightarrow v^2 = \left(\frac{T}{\mu}\right)$$

$$\mu = \frac{T}{v^2} = \frac{100}{(100)^2} = 0.01\,kg/m$$

$$P_{ave} = 2\pi^2 mvr^2 f^2$$

$$= 2(10)(0.01) \times 100 \times (0.5 \times 10^{-3})^2 \times (100)$$

$$= 0.050 \text{ watt} = 50\,mW$$

22. (3)

$$\Delta = \frac{FL}{Ay}$$

$$= \frac{TL}{Ay}$$

$$= \frac{\left(\frac{mv^2}{L} + mg\right)L}{Ay}$$

$$= \frac{\left(\frac{100\times 25}{1} + 1000\right)\times 1}{0.07\times 10^{-4}\times 20\times 10^{9}}$$

$= 25 \times 10^{-3}$m

23. (5)

Using, $PV = nRT$, we have



$$n_0 = \frac{P_1V_1}{R\times 250}$$

$$n' = 0.75\, n_0 + 0.5 n_0$$

$= 1.25\ n_0$ moles

$$P_2 \times 2V_1 = (1.25)\frac{P_1V_1}{R\times 250}\times R\times 2000$$

$$\Rightarrow \frac{P_2}{P_1} = 5$$

24. (3)

We know

$$V_0 = \sqrt{\frac{GM}{r}}\ \&\ g = \frac{GM}{R^2}$$

From energy conservation

$$U_o + K_i = U_f + K_f$$

$$-\frac{GMm}{R} + K_f = -\frac{GMm}{2R} + \frac{1}{2}mv_0^2$$

$$K_i = \frac{GMm}{2R} = \frac{1}{2}m\left(\sqrt{\frac{GM}{2R}}\right)^2$$

$$\Rightarrow K_i \frac{3GMm}{4R}$$

$$\Rightarrow \quad K_i = \frac{3}{4}mgR$$

25. (18)

$r = 1.5$ m, $F = 12t - 3t^2$ N, $I = 4.5$ kgm^2

Torque, $\tau = I\alpha$

$$\alpha = \frac{\tau}{I} = \frac{1.5\times(12t-3t^2)}{4.5} = \frac{1}{3}(12t-3t^2)$$

$$\Rightarrow \alpha = 4t - t^2$$

$$\frac{d\omega}{dt} = 4t - t^2 = \int d\omega = \int\left(4t-t^2\right)_0^t dt$$

$$\omega = 2t^2 - \frac{t^3}{3}$$

For $\omega = 0$,

$$0 = 2t^2 - \frac{t^3}{3} = t^2\left(2 - \frac{t}{3}\right) = 0$$

$t = 0$, 6s

Since, $\omega = \frac{d\theta}{dt} \Rightarrow d\theta = \omega dt$

$$\Rightarrow \int_0^\theta d\theta = \int_0^6\left(2t^2 - \frac{t^3}{3}\right)dt$$

$$\theta = \left[\frac{2t^3}{3} - \frac{t^4}{12}\right] = \left[\frac{t^2}{3}\left(2t - \frac{t^2}{4}\right)\right]_0^6$$

$$= \frac{36}{3}\left(12 - \frac{36}{4}\right) = 36 \text{ rad.}$$

No. of revolutions $= \frac{36}{2\pi} = \frac{18}{\pi}$

$\Rightarrow K = 18$

CHEMISTRY

26. (1)

Reversible adiabatic process is also called on Isoentropic process.

27. (3)

High spin $d^6 : t_{2g}^4 e_g^2$

$CFSE = [-0.4\times 4 + 0.6\times 2]\Delta_0 = -0.4\Delta_0$

Low spin $d^4 : t_{2g}^4 e_g^0$

$CFSE = [-0.4\times 4] = -1.6\Delta_0$

Low spin $d^5 : t_{2g}^5 e_g^0$

$CFSE = [-0.4\times 5]\Delta_0 = -2.0\Delta_0$

High spin $d^7 : t_{2g}^5 e_g^2$

$CFSE = [-0.4\times 5 + 0.6\times 2]\Delta_0 = -0.8\Delta_0$

28. (2)

28. (2)

Millimoles of excess HCl = $24.2 \times 0.1 = 2.42$

Millimoles of HCl reacted with Carbonate

$= 50 \times 0.1 - 2.42 = 5 - 2.42 = 2.58$

$MCO_3 + 2HCl \rightarrow MCl_2 + H_2O + CO_2$

$M_2CO_3 + 2HCl \rightarrow 2MCl + H_2O + CO_2$

Millimoles of Carbonate $= \frac{2.58}{2}$

$$\frac{2.58}{2} = \left(\frac{0.095}{M} \times 10^3\right)$$

M = 73.64 gm (Carbonate will be Li_2CO_3)

29. (2)

Clemmensen reduction

$$>C=O \xrightarrow{Zn\text{-}Hg/HCl} >CH_2$$

30. (1)

$PbSO_4$ is formed

$$\underset{\text{(Lead)}}{Pb(s)} + \underset{\text{(Lead oxide)}}{PbO_2(s)} + 2\underset{\text{(Sulfuric acid)}}{H_2SO_4(aq)} \rightarrow 2\underset{\text{(Lead sulfate)}}{PbSO_4(s)} + 2\underset{\text{(Water)}}{H_2O(l)}$$

31. (2)

High melting point of Cr is attributed to the involvement of greater number of electrons from (n – 1) d i.e. 5 in addition to the ns i.e. 1 electrons in the interatomic metallic bonding.

32. (2)

Step-III is the RDS

Overall ΔH would be negative as aromatisation takes place.

33. (4)

$AlPO_4 \rightleftharpoons Al^{3+} + PO_4^{3-}$

$i = 1 + x = 2$

$\Delta T_b = \text{molality} \times k_b \times i$

$$\therefore \frac{\Delta T_b}{K_b} = 0.02$$

34. (4)

(1) $PhNH_2 + CHCl_3 + KOH \rightarrow PhNC + 3\,KCl + 3H_2O$ (unpleasent smell (Isocyaride test))

$PhNHM_e + CHCl_3 + KOH$ – ve Isocyaide test

(2) $PhNH_2 + NaNO_2 + HCl \rightarrow Ph\overset{\oplus}{N}_2Cl$ (Diazonium reactio)

$PhNHMe + NaNO_3 + HCl \rightarrow Ph{-}N(NO)(Me)$ (Yellow oil)

35. (4)

$$\phi = \frac{hC}{\lambda_0} \therefore 2.4 = \frac{12400}{\lambda_0} \therefore \lambda_0 \approx 516.7 \text{ nm}$$

For PEE, $\lambda \le \lambda_0$. So, 5 sources (A, B, C, D, E) will exhibit photoelectric effect.

36. (4)

W : Phosphorus	Y : Oxygen	X : Sulphur	Z : Chlorine
Electronegativity :	O > Cl > S > P	Catenation :	S > P > O > Cl
Electron Affinity :	Cl > O > S > P	Oxygen exhibits covalency of two only	

37. (4)

Lysine is basic amino acid whereas all others are acidic amino acids.

38. (1)

$H^+ + Fe_{0.9}O + MnO_4^- \rightarrow$

$Mn^{2+} + Fe_2O_3 + H_2O$

$Fe_{0.9}O$

$0.9x - 2 \times 1 = 0$

$$x = \frac{20}{9}$$

$Fe_{0.9}O \rightarrow Fe_2O_3$

$$n_f = \left(3 - \frac{20}{9}\right) \times 0.9 = \frac{7}{10}$$

39. (1)

(I) $$\ln \frac{K_2}{K_1}$$

$$= \frac{E_a}{R}\left[\frac{T_2 - T_1}{T_2 T_1}\right] = \frac{65000}{8.314}\left[\frac{25}{273 \times 298}\right] = 2.4$$

$$\Rightarrow \quad \frac{K_2}{K_1} = 11$$

$T_2 = 298$ K ; $T_1 = 273$ K ,

$E_a = 65000$ J , R = 8.314 J/(mol K)

(II) $$\frac{2.5}{20} = \frac{1}{8} = \left(\frac{1}{2}\right)^n$$

$$\Rightarrow \quad n = 3 \Rightarrow t = 3 \times \frac{0.693}{0.0693} = 30$$

(III) Zero order: $t_{1/2} = \frac{a}{2K_1}$

I order : $t_{1/2} = \frac{0.693}{K_2}$

$\frac{1}{2K_1} = \frac{0.693}{K_2}$

$\Rightarrow \quad \frac{K_2}{K_1} = 2 \times 0.693$

(IV) $t_{1/2} \propto (a)^{1-n}$

$\Rightarrow \quad \frac{480}{240} = \left(\frac{0.0677}{0.136}\right)^{n-1}$

$n = 0$

40. (2)

Based on theory

41. (1)

$H_3X \rightleftharpoons H^+ + H_2X^-$

C 0 0

$C(1-\alpha_1)$ $C\alpha_1$ $C\alpha_1$

$H_2X^- \rightleftharpoons H^+ + HX^{2-}$

$C\alpha_1$ 0 0

$C\alpha_1(1-\alpha_2)$ $C\alpha_2\alpha_1$ $C\alpha_2\alpha_1$

$HX^{2-} \rightleftharpoons H^+ + X^{3-}$

$C\alpha_2\alpha_1$ 0 0

$C\alpha_2\alpha_1(1-\alpha_3)$ $C\alpha_1\alpha_2\alpha_3$ $C\alpha_1\alpha_2\alpha_3$

$[H^+] = (3\times10^{-3}) + \left(3\times10^{-3}\times\frac{1}{3}\right) + 3\times10^{-3}\times0$

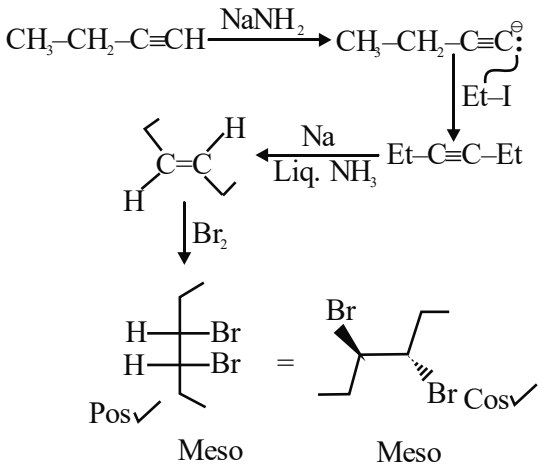
$= 4\times10^{-3}\,M$

$pH = 3 - \log 4 \approx 2.4$

42. (1)

Major product is 3° alcohol formed by reaction of G.R. on ketone

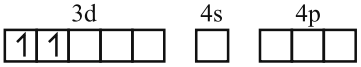
43. (2)



44. (1)

V^{3+} in $[V(H_2O)_6]^{3+}$

3d 4s 4p



$\mu_{Exp} = \sqrt{2(2+2)} = 2.83$ B.M.

45. (3)

$CH_3Cl > CH_3F > CH_3Br$

46. (89)

Mass of organic compound, w = 0.1170 g

mass of AgCl, w_1 = 0.42 g

$AgCl \equiv Cl$

143.5 g 35.5 g

Percentage of chlorine = $\frac{35.5}{143.5} \times \frac{w_1}{w} \times 100 = \frac{35.5}{143.5} \times$

$\frac{0.42}{0.1170} \times 100 \simeq 89\ \%$

47. (15)

$K_{eq} = \frac{K_f}{K_b} = \frac{[B]^2}{[A]}$

$\frac{1.5\times10^{-3}}{K_b} = \left(\frac{100}{10}\right)^2\left(\frac{10}{10^{-5}}\right)$

$K_b = \frac{1.5\times10^{-3}\times100\times10^{-5}}{10^4\times10}$

$= 1.5\times10^{-11}\,L\,mol^{-1}\,s^{-1}$

48. (3)

Equivalent of A1 formed

$= \frac{Q}{F} =$ moles $\times\, n_f$

$Al^{3+} + 3e^- \rightarrow Al(s)$

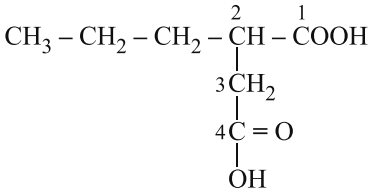
$\frac{0.1\times96.5\times3600\times1000}{96500\times10.8\times100} = mol\times3$

$mol = \frac{0.1\times96.5\times3600\times1000}{96500\times10.8\times100}$

weight of A1 = mol × molar mass

$= \frac{0.1\times96.5\times3600\times1000}{96500\times10.8\times100}\times27 = 3$

49. (4)



50. (1)

$$Sb_2S_3(s) + 3H_2(g) \longrightarrow 2Sb(s) + 3H_2S(g)$$

$$0.01$$

$$0.005$$

$$H_2S + Pb^{2+} \longrightarrow PbS + 2H^+$$

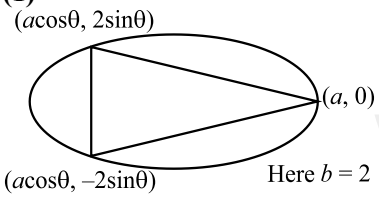
$$PbS = 1.19\text{gm} \Rightarrow \frac{1.19}{239} \approx 0.005 \text{ mole}$$

$$\Rightarrow [H_2] = \left(\frac{0.005}{250}\right); \quad \Rightarrow [H_2S] = \left(\frac{0.005}{250}\right)$$

$$K_c = \left(\frac{0.005}{0.005}\right)^3 = 1$$

MATHEMATICS

51. (1)



$$A = \frac{1}{2}a(1-\cos\theta)(4\sin\theta)$$

$A = 2a\,(1 - \cos\theta)\sin\theta$

$$\frac{dA}{d\theta} = 2a\left(\sin^2\theta + \cos\theta - \cos^2\theta\right)$$

$$\frac{dA}{d\theta} = 0 \Rightarrow 1 + \cos\theta - 2\cos^2\theta = 0$$

$\cos\theta = 1$ (Reject)

OR

$$\cos\theta = -\frac{1}{2} \Rightarrow \theta = \frac{2\pi}{3}$$

$$\frac{d^2A}{d\theta^2} = 2a\left(2\sin^2\theta - \sin\theta\right)$$

$$\frac{d^2A}{d\theta^2} < 0 \text{ for } \theta = \frac{2\pi}{3}$$

$$\text{Now, } A_{\max} = \frac{3\sqrt{3}}{2}a = 6\sqrt{3}$$

$a = 4$

$$\text{Now, } e = \sqrt{\frac{a^2 - b^2}{a^2}} = \frac{\sqrt{3}}{2}$$

52. (1)

$\because f(x+y) = f(x)\cdot f(y) \forall x, y \in R$

Given $f(1) = 3$

$\Rightarrow f(x) = 3^x \Rightarrow f(i) = 3^i$

$$\Rightarrow \sum_{i=1}^{n} f(i) = 363$$

$$\Rightarrow 3 + 3^2 + 3^3 + \ldots + 3^n = 363$$

$$\Rightarrow \frac{3(3^n - 1)}{3 - 1} = 363$$

$$\Rightarrow 3^n - 1 = \frac{363 \times 2}{3} = 242$$

$$\Rightarrow 3^n = 243 = 3^5$$

Hence, $n = 5$

53. (4)

$$2\{(x-p)(x-q) + (p-q)(p-x-q+x\}$$

$$= (p-q)^2 + (x-p)^2 + (x-q)^2$$

$$\Rightarrow \quad 2(x-p)(x-q) + (p-q)^2$$

$$= (x-p)^2 + (x-q)^2$$

$$\Rightarrow \quad 2x^2 - 2x(p+q) + 2pq + p^2 + q^2 - 2pq$$

$$= 2x^2 + p^2 + q^2 - 2x(p+q)$$

It is an identify so infinite values of x are possible.

54. (4)

With the first 20 natural numbers, there are 19 pairs of consecutive numbers.

Let $a = n$, $b = n + 1$ so that $n = 1, 2, 3, \ldots, 19$.

Now, $a^2 + b^2 + a^2b^2 = n^2 + (n+1)^2 + n^2\,(n+1)^2$

$= n^4 + 2n^3 + 3n^2 + 2n + 1$

$= (n^2 + n + 1)^2$

$$\therefore \sqrt{a^2 + b^2 + a^2b^2} = n^2 + n + 1 = n(n+1) + 1$$

Which is always odd because $n(n+1)$ is always even.

So, probability = 1

55. (2)

$$100\tan\left(3\tan^{-1}\frac{1}{2}+2\tan^{-1}\left(\frac{1}{2}\right)+2\tan^{-1}(2)\right)+$$

$$4\sqrt{2}\tan\left(\frac{\tan^{-1}(2\sqrt{2})}{2}\right)$$

$$\Rightarrow 50\left(\frac{1}{2}\right)+4\sqrt{2}\tan\alpha$$

Where $2\alpha=\tan^{-1}2\sqrt{2}$

$$\Rightarrow \frac{2\tan\alpha}{1-\tan^2\alpha}=2\sqrt{2} \quad \ldots\ldots\ldots(i)$$

$$\Rightarrow (2\sqrt{2}\tan\alpha-2)(\tan\alpha-\sqrt{2})=0$$

$\Rightarrow (\tan\alpha=\sqrt{2}$ does not satisfy (i))

$$\Rightarrow 50+4\sqrt{2}\frac{1}{\sqrt{2}}=54$$

56. (1)

(I) We have

$$=\int\frac{(\cos 5x+\cos 4x)}{1-2\cos 3x}dx$$

$$=\int\frac{(\cos 5x+\cos 4x)\sin 3x}{\sin 3x-\sin 6x}dx$$

$$=\int\frac{\left(2\cos\frac{9x}{2}\cos\frac{x}{2}\right)\sin 3x}{-2\cos\frac{9x}{2}\sin\frac{3x}{2}}dx$$

$$=\int\frac{-\cos\frac{x}{2}\left(2\sin\frac{3x}{2}\cos\frac{3x}{2}\right)}{\sin\frac{3x}{2}}dx$$

$$=-\int(\cos 2x+\cos x)dx$$

$$=-\left(\frac{1}{2}\sin 2x+\sin x\right)+c$$

$$=-\frac{1}{2}(2\sin x+\sin 2x)+c$$

Answer (I)→(R)

(II) We have

$$=\int\frac{(\cos 7x-\cos 8x)\sin 5x}{\sin 5x+\sin 10x}dx$$

$$=\int\frac{\left(2\sin\frac{15x}{2}\sin\frac{x}{2}\right)\left(2\sin\frac{5x}{2}\cos\frac{5x}{2}\right)}{2\sin\frac{15x}{2}\cdot\cos\frac{5x}{2}}dx$$

$$=\int 2\sin\frac{x}{2}\sin\frac{5x}{2}dx$$

$$=\int(\cos 2x-\cos 3x)dx$$

$$=\frac{1}{2}\sin 2x-\frac{1}{3}\sin 3x+c$$

Answer (II)→(S)

(III) We have

$$I=\int\tan^3 2x\sec 2xdx$$

$$=\int\tan^2 2x(\sec 2x\tan 2x)dx$$

$$=\int\left(\sec^2 2x-1\right)\sec 2x\tan 2xdx$$

Put $t=\sec 2x\$$, so that $\frac{dt}{2}$

$$=(\sec 2x\tan 2x)dx$$

Therefore,

$$=\frac{1}{2}\int\left(t^2-1\right)dt$$

$$=\frac{1}{2}\left(\frac{1}{3}t^3-t\right)+c$$

$$=\frac{1}{6}\sec^3 2x-\frac{1}{2}\sec 2x+c$$

Answer (III)→(P)

(IV) We have

$$I=\int\sqrt{2+\sin 3x}\cos 3xdx$$

Put $2+\sin 3x=t$.

Therefore $3\cos 3xdx=dt$. Now

$$=\int\sqrt{t}\cdot\frac{1}{3}dt=\frac{2}{9}(t)^{3/2}+c$$

$$=\frac{2}{9}(2+\sin 3x)^{3/2}+c$$

Answer (IV)→(Q)

57. (2)

Before rotation $\vec{a}=2p\hat{i}+\hat{j}$

after rotation $\vec{a}=(p+1)\hat{i}'+\hat{j}'$

Since length of vector remains unaltered

$$\sqrt{4p^2+1}=\sqrt{(p+1)^2+1}$$

$4p^2=(p+1)^2 \Rightarrow p+1=\pm 2p$

$p=1$ or $-\frac{1}{3}$

58. (2)

Let $Z=x+iy, x\in, y\in R$

$\therefore \quad x-iy=i\,(x^2-y^2+(2xy)i+x)$

$x=-2xy$ .....(i)

$-y=-y^2+x^2+x$ .....(ii)

$\Rightarrow \quad x=0, y=-\frac{1}{2}$ (from (i) and (ii), we have)

When $x=0$, then $y=0, 1$

When $y=-\frac{1}{2}$, then $x=\frac{1}{2},-\frac{3}{2}$

$\therefore \quad Z=0,0+i,\frac{1}{2}-\frac{i}{2},-\frac{3}{2}-\frac{i}{2}$

59. (1)

$AB=O$

$$\Rightarrow\begin{bmatrix}\cos^2\theta\cos^2\phi+\sin\theta\sin\phi\cos\theta\cos\theta & \cos^2\theta\cos\phi\sin\phi+\sin^2\phi\cos\theta\sin\theta\\ \cos^2\phi\cos\theta\sin\theta+\sin^2\theta\cos\phi\sin\phi & \cos\theta\sin\theta\cos\phi\sin\phi+\sin^2\theta\sin^2\phi\end{bmatrix}$$

$$=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

$$\Rightarrow\begin{bmatrix}\cos\theta\cos\phi\cos(\theta-\phi) & \cos\theta\sin\phi(\cos\theta-\phi)\\ \cos\phi\sin\theta\cos(\theta-\phi) & \sin\theta\sin\phi\cos(\theta-\phi)\end{bmatrix}$$

$$=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

$\Rightarrow \cos(\theta-\phi)=0\Rightarrow \theta-\phi=(2n+1)\frac{\pi}{2}$

60. (1)

Required number of ways $= {}^5C_1\,D_4+{}^5C_0\,D_5$

$$=5\times 4!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}\right)$$

$$+1\times 5!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}\right)=89$$

61. (1)

Let

$$f_n(x)=\begin{cases}\frac{x^n}{n!} & \text{for } 0\le x\le\frac{1}{2}\\ \frac{(1-x)^n}{n!} & \text{for } \frac{1}{2}<x\le 1\end{cases}$$

Therefore

$$I_n=\int_0^{1/2}\frac{x^n}{n!}dx+\int_{1/2}^1\frac{(1-x)^n}{n!}dx$$

$$=\frac{1}{(n+1)!}\left(\frac{1}{2}\right)^{n+1}-\left[0-\frac{1}{(n+1)!}\left(\frac{1}{2}\right)^{n+1}\right]$$

$$=2\frac{(1/2)^{n+1}}{(n+1)!}$$

Hence

$$\sum_{n=1}^{\infty}I_n=2\left[\frac{(1/2)^2}{2!}+\frac{(1/2)^3}{3!}+\frac{(1/2)^4}{4!}+\cdots+\infty\right]$$

$$=2\left[\left(1+\frac{(1/2)}{1!}+\frac{(1/2)^2}{2!}+\frac{(1/2)^3}{3!}+\cdots+\infty\right)-1-\frac{1}{2}\right]$$

$$=2\left[e^{1/2}-\frac{3}{2}\right]=2\sqrt{e}-3$$

62. (2)

$$\omega^n=\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)^n$$

$$=\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)^n$$

$$=\cos\frac{2n\pi}{3}+i\sin\frac{2n\pi}{3}$$

Now $\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)^n=\frac{(-1)^n}{2^n}(1-i\sqrt{3})^n$

$$=\frac{(-1)^n}{2^n}[1+{}^nC_1(-i\sqrt{3})+{}^nC_2(-i\sqrt{3})^2+nC_3$$

$(-i\sqrt{3})^3\ \ldots]$

$$=\frac{(-1)^n}{2^n}[(1-3^nC_2\ldots)-i\sqrt{3}({}^nC_1-3^nC_3+3^2\,{}^nC_5-3^3$$

${}^nC_7\ldots)]$

Equating the real and imaginary parts, we get

$$S=(-1)^{n+1}\frac{2^n}{\sqrt{3}}\sin\frac{2n\pi}{3}$$

63. (4)

$$\frac{x}{1}=\frac{y}{2}=\frac{z}{3}=r \qquad \ldots(1)$$

$$\frac{x-1}{3}=\frac{y-2}{-1}=\frac{z-3}{4} \qquad \ldots(2)$$

$$\frac{x+k}{3}=\frac{y-1}{2}=\frac{z-2}{h} \qquad \ldots(3)$$

$\because$ $P(r, 2r, 3r)$ lies on (2)

$$\therefore\ \frac{r-1}{3}=\frac{2r-2}{-1}=\frac{3r-3}{4}\Rightarrow r=1$$

$\therefore$ point of intersection of (1) and (2) be (1,2,3).

(1, 2, 3) will also satisfy (3) as the lines are concurrent

$$\therefore\ \frac{1+k}{3}=\frac{1}{2}=\frac{1}{h}\Rightarrow h=2;\ k=\frac{1}{2}$$

64. (2)

Let

$$T_n=\sqrt{1+\frac{1}{n^2}+\frac{1}{(n+1)^2}}=\sqrt{\frac{n^2(n+1)^2+(n+1)^2+n^2}{n^2(n+1)^2}}$$

Then

$$T_n=\frac{n^2+n+1}{n(n+1)}=1+\frac{1}{n(n+1)}=1+\frac{1}{n}-\frac{1}{n+1}$$

Therefore,

$$x=\sum_{n=1}^{2009}T_n=\sum_{n=1}^{2009}\left(1+\frac{1}{n}-\frac{1}{n+1}\right)$$

$$=2009+\left(1-\frac{1}{2009+1}\right)$$

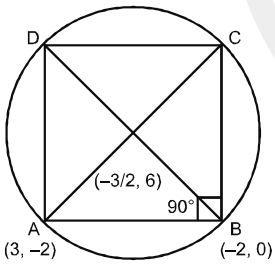
$$=\frac{(2010)^2-1}{2010}$$

$$=\frac{2011\cdot 2009}{2010}$$

Therefore, $\frac{2010x-2009}{2010}=2009$

65. (1)

The centre $\left(-\frac{3}{2},-6\right)$ is mid-poins of diameter AC and BD

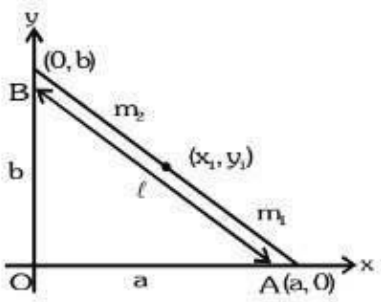


$\therefore$ C is $(p, q) = (-6, -10)$

$\therefore$ D is $(r, s) = (-1, -12)$

66. (3)

Let (x_1, y_1) be the point that divide the rod $AB = \ell$, in the ratio $m_1 : m_2$, and $OA = a, OB = b$ say



$\therefore$ $a^2+b^2=t^2$

Now $x_1=\left(\frac{m_2a}{m_1+m_2}\right)\Rightarrow a=\left(\frac{m_1+m_2}{m_2}\right)x_1$

$y_1=\left(\frac{m_1b}{m_1+m_2}\right)\Rightarrow b=\left(\frac{m_1+m_2}{m_1}\right)y_1$

putting these values in (i)

$$\frac{(m_1+m_2)^2}{m_2^2}x_1^2+\frac{(m_1+m_2)^2}{m_1^2}y_1^2=\ell^2$$

$\therefore$ Locus of (x_1, y_1) is

$$m_1^2x^2+m_2^2y^2=\left(\frac{m_1m_2\ell}{m_1+m_2}\right)^2$$

67. (4)

$1\le\cos^{-1}\sin^{-1}\tan^{-1}x<2$

$\Rightarrow\cos2<\sin^{-1}\tan^{-1}x\le\cos1$

$\Rightarrow\sin(\cos2)<\tan^{-1}x\le\sin\cos1$

$\Rightarrow\tan\sin(\cos2)<x\le\tan\sin\cos1$

68. (4)

We have, $\frac{\sqrt{x}dx+\sqrt{y}dy}{\sqrt{x}dx-\sqrt{y}dy}=\sqrt{\frac{y^3}{x^3}}$

$$\Rightarrow\frac{d\left(x^{3/2}\right)+d\left(y^{3/2}\right)}{d\left(x^{3/2}\right)-d\left(y^{3/2}\right)}=\frac{y^{3/2}}{x^{3/2}}$$

$\Rightarrow \frac{du+dv}{du-dv}=\frac{v}{u}$, where $u=x^{3/2}$ and $v=y^{3/2}$

$\Rightarrow$ $udu+udv=vdu-vdv$

$\Rightarrow$ $udu+vdv=vdu-udv$

$$\Rightarrow\frac{udu+vdv}{u^2+v^2}=\frac{vdu-udv}{u^2+v^2}$$

$$\Rightarrow\frac{d\left(u^2+v^2\right)}{u^2+v^2}=-2d\tan^{-1}\left(\frac{v}{u}\right)$$

On integrating , we get

$$\log\left(u^2+v^2\right)=-2\tan^{-1}\left(\frac{v}{u}\right)+c'$$

$$\Rightarrow\frac{1}{2}\log\left(x^3+y^3\right)+\tan^{-1}\left(\frac{y}{x}\right)^{3/2}=\frac{c'}{2}$$

$$\Rightarrow\frac{1}{2}\log\left(x^3+y^3\right)+\tan^{-1}\left(\frac{y}{x}\right)^{3/2}+c=0$$

69. (4)

Since $AB = B$ and $BA = A$

A and B both are idempotent

$(A-B)^2 = A^2 - AB - BA + B^2$

$= A - B - A + B = 0$

$A - B$ is nilpotent

70. (3)

It is given that

$$xy = ax^2 + \frac{b}{x}$$

Differentiating both sides w.r.t. x we get

$$y + x\frac{dy}{dx} = 2ax - \frac{b}{x^2}$$

$$\Rightarrow x^2 y + x^3 \frac{dy}{dx} = 2ax^3 - b$$

Again, differentiating both sides w.r.t. x we get

$$2yx + x^2\frac{dy}{dx} + 3x^2\frac{dy}{dx} + x^3\frac{d^2y}{dx^2} = 6ax^2$$

$$\Rightarrow 2xy + 4x^2\frac{dy}{dx} + x^3\frac{d^2y}{dx^2} = 6ax^2$$

$$\Rightarrow 2ax^2 + \frac{2b}{x} + 4x^2\frac{dy}{dx} + x^3\frac{d^2y}{dx^2} = 6ax^2$$

$$\left(\because xy = ax^2 + \frac{b}{x}\right)$$

$$\Rightarrow 4x^2\frac{dy}{dx} + x^3\frac{d^2y}{dx^2} = 4ax^2 - \frac{2b}{x}$$

$$\Rightarrow 4x\frac{dy}{dx} + x^2\frac{d^2y}{dx^2} = 4ax - \frac{2b}{x^2} = \frac{4ax^3 - 2b}{x^2}$$

$$= \frac{2\left(2ax^3 - b\right)}{x^2} = \frac{2}{x^2}\left(x^2 y + x^3\frac{dy}{dx}\right) = 2y + 2x\frac{dy}{dx}$$

Therefore, $2x\frac{dy}{dx} + x^2\frac{d^2y}{dx^2} = 2y$

$$x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = \frac{2y}{x}$$

71. (3)

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} f(x+6) = \lim_{x\to\infty} f(x+7) = m$$

then from the given condition

$$m = \frac{1}{3}\left[m + \frac{6}{m}\right]$$

$\Rightarrow 3m^2 = m^2 + 6$

$\Rightarrow 2m^2 = 6$

$\Rightarrow m = \sqrt{3}$

so, $\lim_{x\to\infty} f(x) = \sqrt{3}$

so the value of $m = 3$

72. (260)

Solution: Observe that the element in the /th row and jth column is $(i-1)8 + j$. Let $a_{1/j}, a_{2h}, \ldots, a_{8j_8}$ be the numbers so selected where $j_1, j_2, \ldots, j_8$ are different. Therefore

$$\sum_{i=1}^{8} a_{ij} = \sum_{i=1}^{8}\left[(i-1)8 + j_i\right]$$

$= 8(0+1+2+3+\cdots+7) + (1+2+3+\cdots+8)$

$= 8 \times 28 + 36$

$= 224 + 36$

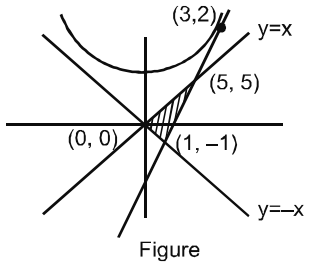
$= 260$

73. (5)

$\left.\frac{dy}{dx}\right|_{(3,2)} = \frac{3}{2}$. Tangent $y = \frac{3x}{2} - \frac{5}{2}$

$y = x, y = \frac{3x}{2} - \frac{5}{2}$

$\Rightarrow$ (5, 5)



Figure

$y = -x, y = \frac{3x}{2} - \frac{5}{2} \Rightarrow (1, -1)$

closed figure formed is right angled triangle. Its area is $\frac{1}{2}(\sqrt{2})\ (5\sqrt{2}) = 5$

74. (3)

variance $(x_i - 4) = \text{var}(x_i) = \frac{44}{11} - \left(\frac{11}{11}\right)^2 = 3$

75. (6)

$y = x^2 + 3x$ …(i)

$y = 5 - x$ …(ii)

Solve these equation $A(1, 4)$, $B(-5, 10)$

$AB = \sqrt{36 + 36} = 6\sqrt{2}$